

FNB Mega Fest 2026

Concert Ticket Booking System

Project Documentation

1. Introduction

The **Concert Ticket Booking System** is a console-based Python application developed as part of my **FNB App Academy 2026** learning journey.

Although the original challenge focused on practicing basic Python concepts such as variables, input, and output, I expanded the scope of the project to create a more realistic software application. My goal was to challenge myself by applying object-oriented programming principles, data structures, input validation, and modular programming while still covering the concepts introduced in the academy.

This project represents the first step in a series of increasingly advanced Python applications that will be developed throughout my learning journey.

2. Project Scenario

An event management company is preparing to host the **FNB Mega Fest 2026**, a large music festival featuring several well-known artists.

The organizers require a digital booking system to replace manual ticket bookings. Customers should be able to purchase one or more tickets, select the artist they are most excited to see, and receive a unique ticket number upon successful booking.

The company also wants to collect customer preferences so that, in future festivals, the booking data can be analysed to determine which artists are most anticipated. This information will later support business intelligence dashboards and predictive machine learning models.

You have been hired as the software developer responsible for designing and implementing the first version of this booking system.

3. Project Objectives

The primary objectives of this project are to:

- Develop a menu-driven Python application.
- Apply Object-Oriented Programming (OOP).

- Practice creating reusable functions.
 - Implement input validation.
 - Generate unique ticket numbers.
 - Store customer bookings efficiently.
 - Simulate a real-world booking system.
 - Create a project that can later be expanded into a full software solution.
-

4. Functional Requirements

The system must:

- Display a user-friendly main menu.
 - Allow customers to purchase one or multiple tickets.
 - Collect customer information including:
 - Name
 - Age
 - Email Address
 - Favourite Artist
 - Display the available performing artists.
 - Generate a unique ticket number for every booking.
 - Prevent invalid user input.
 - Store booking information using appropriate Python data structures.
 - Display all existing bookings.
 - Allow users to exit the application safely.
-

6. Current Features

The current version of the system includes:

Ticket Booking

Users can purchase one or multiple tickets.

Customer Information Collection

The system records:

- Customer Name
- Age
- Email Address
- Favourite Artist

Artist Selection

Customers select their preferred artist from a predefined list.

Ticket ID Generation

Every booking automatically receives a randomly generated unique ticket number.

Input Validation

The application validates:

- Menu selections
- Number of tickets
- Customer age
- Artist selection

Booking Storage

Bookings are stored using a Python dictionary where the ticket number acts as the unique key.

Booking Viewer

Users can display all bookings currently stored in the system.

7. Future Improvements

This project has been intentionally designed so it can grow into a much larger software system.

Planned improvements include:

Version 2

- Search bookings using Ticket ID
- Edit bookings
- Cancel bookings

Version 3

- Save bookings to CSV
- Load bookings from CSV

Version 4

- SQLite Database Integration
- Persistent data storage
- CRUD operations

Version 5

- Revenue tracking
- Ticket pricing
- VIP and General Admission tickets
- Seat allocation

Version 6

- Data Analysis using Pandas
- Booking statistics
- Customer demographics
- Artist popularity reports

Version 7

- Power BI Dashboard
- Interactive visualisations
- Booking trends
- Revenue reports

Version 8

Machine Learning Integration

Using historical booking data to predict:

- Most anticipated artist
- Expected attendance
- Ticket demand

- Customer preferences
-

8. Learning Outcomes

This project demonstrates my ability to:

- Design a structured Python application.
- Apply object-oriented programming principles.
- Build reusable and maintainable code.
- Validate user input effectively.
- Work with collections such as lists and dictionaries.
- Solve real-world software problems using Python.

More importantly, this project serves as the foundation for future projects involving databases, web development, data analytics, business intelligence, and machine learning.

9. Conclusion

The **Concert Ticket Booking System** marks the beginning of my **FNB App Academy 2026** journey.

Rather than completing only the basic exercise, I expanded the challenge into a more realistic software application to reinforce my understanding of Python programming and software design principles.

As I continue through the academy, this project will evolve alongside my skills, eventually incorporating databases, analytics, dashboards, and artificial intelligence to demonstrate the complete software development lifecycle.
